Gaussian Rigidity from Symmetrized Moment Generating Functions: Sharp Reflection Stability and Chi-Square Applications

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Abstract

We study recovery of a Gaussian law from its symmetrized moment generating function on a fixed real interval. Let X be centered with variance one, and suppose that its moment generating function is finite on $[-\rho, \rho]$. If

$$\sup_{|s|\le\rho} \left|\log M_X(s) + \log M_X(-s) - s^2\right| \le \Delta,$$

then, for sufficiently small Δ,

$$d_{\mathrm{K}}(X, \mathcal{N}(0,1)) \le C_\rho \sqrt{\frac{\log\log(1/\Delta)}{\log(1/\Delta)}}.$$

For every fixed $\rho > 0$, a symmetric Laguerre perturbation attains this order while its density remains uniformly comparable with the Gaussian density. Thus the logarithmic rate is necessary even in the symmetric case. The proof combines high-order moment comparison, truncation and a growing zero-free disk for one characteristic-function factor. We also prove an exact characterization for chi-square positive-semidefinite quadratic forms under coordinatewise reflected-MGF dominance. Spherical averaging connects this condition with the sample variance and yields a stability result from Kolmogorov distance to the chi-square law under a fixed exponential-moment bound.

Keywords: Gaussian characterization; symmetrized moment generating function; chi-square quadratic forms; sample variance; Cramér stability; zero-free characteristic functions

MSC Classification: 60E05 , 60E10 , 62E10

1 Introduction and main results

Let X' be an independent copy of a real random variable X. Wherever the moment generating functions are finite,

$$M_{X-X'}(s) = M_X(s)M_X(-s).$$

This product determines the symmetrized law, but it need not determine the law of X: replacing X by $-X$ leaves the product unchanged. The Gaussian case is rigid. If $X - X'$ is Gaussian, Cramér's decomposition theorem implies that X is Gaussian [1]. We ask for a quantitative version when the reflected product is known to be approximately Gaussian only on a fixed real interval. The aim is to recover one factor and to determine the best possible dependence on the local error.

For a centered variance-one random variable X, write $M_X(s) = \mathbb{E}e^{sX}$ and define

$$R_X(s) := \log M_X(s) + \log M_X(-s) - s^2, \qquad \Delta_\rho(X) := \sup_{|s|\le\rho} |R_X(s)|. \tag{1}$$

We use the Kolmogorov distance

$$d_{\mathrm{K}}(X, Z) := \sup_x \left|\mathbb{P}(X \le x) - \mathbb{P}(Z \le x)\right|,$$

and denote the standard Gaussian density and distribution function by ϕ and Φ.

1.1 Sharp recovery from local reflected information

The main result requires neither symmetry nor a sign condition on R_X.

Theorem 1.1 (Sharp reflection stability) *Fix $\rho > 0$. There exist constants $C_\rho < \infty$ and $\Delta_0(\rho) \in (0, e^{-e})$ such that the following holds. Let X satisfy*

$$\mathbb{E}X = 0, \qquad \mathbb{E}X^2 = 1, \qquad M_X(s) < \infty \quad (|s| \le \rho),$$

and suppose that $\Delta_\rho(X) \le \Delta$ for some $0 < \Delta \le \Delta_0(\rho)$. Then

$$d_{\mathrm{K}}(X, \mathcal{N}(0,1)) \le C_\rho \sqrt{\frac{\log\log(1/\Delta)}{\log(1/\Delta)}}.$$

The local hypothesis already controls the exponential moment used in the proof. Since $\mathbb{E}X = 0$, Jensen's inequality gives $M_X(s) \ge 1$. Consequently,

$$\mathbb{E}e^{\rho|X|} \le M_X(\rho) + M_X(-\rho) \le 2e^{\rho^2+\Delta}. \tag{2}$$

No additional uniform moment bound is assumed. If $\Delta_\rho(X) = 0$, uniqueness of the MGF and Cramér's theorem give $X \sim \mathcal{N}(0, 1)$.

The rate in Theorem 1.1 is attained for every fixed observation radius. The lower-bound family is symmetric and has Gaussian-comparable densities.

Theorem 1.2 (Symmetric Laguerre lower bound) *Fix $\rho > 0$. There are constants $c, c_-, c_+ > 0$ and centered variance-one random variables W_m with symmetric densities f_m such that, for all sufficiently large m,*

$$c_-\phi(x) \leq f_m(x) \leq c_+\phi(x), \qquad R_{W_m}(s) \geq 0 \quad (s \in \mathbb{R}).$$

With $\Delta_m := \sup_{|s|\leq\rho} R_{W_m}(s)$, one has $\Delta_m \to 0$ and

$$\log \frac{1}{\Delta_m} = m \log m + O_\rho(m), \qquad d_{\mathrm{K}}(W_m, \mathcal{N}(0,1)) \geq cm^{-1/2}.$$

In particular, for some $c'_\rho > 0$ and all sufficiently large m,

$$d_{\mathrm{K}}(W_m, \mathcal{N}(0,1)) \geq c'_\rho \sqrt{\frac{\log\log(1/\Delta_m)}{\log(1/\Delta_m)}}.$$

For this family, $M_{W_m}(-s) = M_{W_m}(s)$, so the real-axis MGF factor is the positive square root of the reflected product. The lower bound therefore concerns the passage from local transform information to a global probability metric, not only the loss of Fourier phase. In the nonsymmetric case, factor recovery must also deal with the identity $\varphi_{X-X'}(t) = |\varphi_X(t)|^2$.

1.2 Relation to decomposition and zero-free methods

Quantitative Gaussian decomposition has been studied since Sapogov [2]. The monograph of Linnik and Ostrovskii [3] develops the analytic theory of decompositions, and Golinskii and Chistyakov [4] give order-sharp stability estimates in the Lévy metric. Classical stability estimates in this setting start from global proximity of a convolution to a Gaussian law. For identically distributed summands, Bobkov, Chistyakov, and Götze [5] obtain stronger Kolmogorov estimates than those available for general decompositions. Their equal-summand problem concerns $X + X'$, whereas the reflected product here corresponds to $X - X'$. More importantly, our hypothesis controls this product only on a fixed real interval, rather than controlling the convolution in a probability metric.

A second connection is with Marcinkiewicz's theorem, which states that a characteristic function of the form e^P, with P a polynomial, must be Gaussian or degenerate [6]. Golinskii [7] studies stability when e^P is close to a characteristic function. Eremenko and Fryntov [8] derive Gaussian approximation from a zero-free vertical strip and a global growth condition. Dinh, Ghosh, Tran, and Tran [9, Theorem 2.2] require only a zero-free disk and a growth bound on that disk. Related results for probability generating polynomials connect the locations of their zeros with central limit theorems [10–12].

In our proof the zero-free region is not assumed. Local reflected-MGF control first gives accurate moments through order $2m$, where $m \asymp \log(1/\Delta)/\log\log(1/\Delta)$. Truncation at $N \asymp \sqrt{m}$ then yields a zero-free disk of radius $R \asymp \sqrt{m}$ for a single factor. If one uses only the bound $|\varphi_N(z)| \leq e^{NR}$ in the disk theorem of Dinh et al., the resulting estimate is $O((1 + \log R)/R)$. A direct Carathéodory coefficient estimate adapted to $N \asymp R$ gives $O(R^{-1})$ instead, as proved in Lemma 3.7. This is the scale required for Theorem 1.1.

Symmetric perturbations with densities comparable to the Gaussian also occur in the strong-distance counterexamples of Bobkov, Chistyakov, and Götze [5]. The

Laguerre perturbation in Theorem 1.2 is chosen to cancel the first $2m-1$ moments and to give an explicit, nonnegative reflected defect of factorial size. These two features relate the local MGF error to the Kolmogorov lower bound at the same scale as the upper theorem.

1.3 Quadratic forms and the sample variance

Quadratic-form characterization provides a source of the reflected quantity. The classical sample-variance problem asks whether, for a fixed $n \geq 3$, a chi-square law for $\sum_i (X_i - \bar{X})^2$ forces an i.i.d. sample to be Gaussian. The unrestricted question is recorded in Kagan, Linnik, and Rao [13, p. 466] and discussed by Ejsmont and Lehner [14], whose main results concern the free-probability analogue. Ruben [15] proves the classical characterization for symmetric laws and for $n = 2$. A chi-square hypothesis at two distinct sample sizes is treated by Ruben [16] and Bondesson [17]. Golikova and Kruglov [18] obtain a sample-variance characterization for independent infinitely divisible variables with a common mean.

For more general quadratic forms, see Ruben [19] and the characterization and stability results for i.i.d. symmetric variables in Christoph, Prohorov, and Ulyanov [20]. We impose a different condition, namely local reflected-MGF dominance. The following theorem does not require the coordinates to have a common distribution.

Theorem 1.3 (Exact PSD rigidity) *Let $X = (X_1, \ldots, X_d)$ have independent coordinates with $\mathbb{E}X_i = 0$ and $\mathrm{Var}(X_i) = \sigma_i^2$. Let $A = A^\top \succeq 0$ have rank $r \geq 1$, and suppose that*

$$X^\top A X \sim \chi_r^2.$$

For every active coordinate i, meaning $A_{ii} > 0$, assume that

$$M_i(s) M_i(-s) \geq e^{\sigma_i^2 s^2}$$

for all s in a neighborhood of the origin. Then every active X_i is Gaussian, with law $\mathcal{N}(0, \sigma_i^2)$.

The chi-square hypothesis itself supplies a square-exponential moment for each active coordinate, so MGF existence is not an additional assumption in this theorem. Among laws with a local MGF, infinite divisibility implies the dominance condition by the Lévy–Khintchine formula [21]. The converse fails, and dominance does not require symmetry. Both points are made explicit in Remark 2.1.

Taking $A = I_n - n^{-1}\mathbf{1}\mathbf{1}^\top$ gives the sample-variance characterization under dominance. In the quantitative version, a spherical Laplace observable transfers information about the quadratic form to the reflected defect. Under a fixed exponential envelope, Kolmogorov distance to χ_{n-1}^2 controls this observable and hence the law of X; see Corollary 5.4. This application retains the dominance assumption and does not settle the unrestricted sample-variance conjecture. We do not assert that its modulus is sharp.

Organization. Section 2 proves the exact quadratic-form theorem and introduces spherical averaging. Section 3 proves the reflection upper bound, and Section 4 gives the matching lower bound and the total-variation obstruction. Section 5 develops the sample-variance application. The appendices contain the parameter choices and the Laguerre estimate.

2 Exact rigidity for chi-square quadratic forms

The proof uses the chi-square law to identify a spherical average of the product of the coordinate MGFs. Jensen's inequality and reflected dominance then force each coordinate defect to vanish. We first prove the theorem and then discuss the scope of its dominance assumption.

Proof of Theorem 1.3 Write $A = B^\top B$, where $B \in \mathbb{R}^{r\times d}$ has full row rank, and let b_i be the ith column of B. Thus $A_{ii} = \|b_i\|^2$.

We first note that the chi-square hypothesis automatically gives the MGF regularity needed below. Fix an active coordinate i, so $A_{ii} > 0$, and let

$$Q := X^\top AX \sim \chi_r^2.$$

Independence and centering imply

$$\mathbb{E}(Q \mid X_i) = A_{ii}X_i^2 + \sum_{j\neq i} A_{jj}\sigma_j^2.$$

For every $0 < c < 1/2$, conditional Jensen therefore yields

$$\exp\left(cA_{ii}X_i^2 + c\sum_{j\neq i} A_{jj}\sigma_j^2\right) \le \mathbb{E}(e^{cQ} \mid X_i).$$

Taking expectations and using the chi-square MGF gives

$$\mathbb{E}e^{cA_{ii}X_i^2} < \infty. \tag{3}$$

Consequently $M_i(s) < \infty$ for every $s \in \mathbb{R}$, by Young's inequality

$$sx \le cA_{ii}x^2 + \frac{s^2}{4cA_{ii}}.$$

If V is uniform on S^{r-1}, write

$$\Psi_r(z) := \mathbb{E}_V e^{zV_1}.$$

The transform is well defined for all $z \in \mathbb{C}$ because V_1 is bounded. Spherical averaging gives

$$\mathbb{E}_V \prod_{i=1}^d M_i\big(t\langle b_i, V\rangle\big) = \mathbb{E}_{X,V} e^{t\langle BX,V\rangle} = \mathbb{E}\Psi_r\big(t\sqrt{X^\top AX}\big). \tag{4}$$

Since $X^\top AX \sim \chi_r^2$, the final quantity in (4) equals

$$\mathbb{E}_{G,V} e^{t\langle G,V\rangle} = e^{t^2/2}, \qquad G \sim \mathcal{N}(0, I_r). \tag{5}$$

Only active coordinates enter the product, because $A_{ii} = 0$ implies $b_i = 0$.

For each active coordinate, let the local dominance hold on $|s| < \rho_i$. Since there are finitely many active coordinates, reducing to a common radius $\rho_* := \min_i \rho_i > 0$ makes all subsequent expressions valid for sufficiently small $|t|$.

Define

$$C_i(s) := \log M_i(s) - \frac{\sigma_i^2 s^2}{2}$$

and

$$H_t(v) := \sum_{i=1}^d C_i\big(t\langle b_i, v\rangle\big) + \frac{t^2}{2}\left(\sum_{i=1}^d \sigma_i^2\langle b_i, v\rangle^2 - 1\right). \tag{6}$$

The quadratic terms cancel inside the exponential, so (5) implies

$$\mathbb{E}_V e^{H_t(V)} = e^{-t^2/2}\mathbb{E}_V \prod_i M_i(t\langle b_i, V\rangle) = 1.$$

Jensen's inequality therefore gives $\mathbb{E}_V H_t(V) \le 0$.

The mean of the quadratic correction in (6) is zero. Indeed,

$$\mathbb{E}_V \sum_i \sigma_i^2 \langle b_i, V\rangle^2 = \frac{1}{r}\sum_i \sigma_i^2 \|b_i\|^2 = \frac{1}{r}\mathbb{E}(X^\top AX) = 1,$$

where the last equality uses the mean of χ_r^2. Since V and $-V$ have the same law, coordinatewise dominance now gives

$$\begin{aligned} 2\mathbb{E}_V H_t(V) &= \sum_{i=1}^d \mathbb{E}_V \left[\log M_i(t\langle b_i, V\rangle) + \log M_i(-t\langle b_i, V\rangle) - \sigma_i^2 t^2 \langle b_i, V\rangle^2\right] \\ &= \sum_{i=1}^d \mathbb{E}_V R_i\big(t\langle b_i, V\rangle\big) \ge 0, \end{aligned}$$

where $R_i(s) = \log M_i(s) + \log M_i(-s) - \sigma_i^2 s^2$. Hence $\mathbb{E}_V H_t(V) = 0$, and every nonnegative summand in the last display has zero expectation.

For an active coordinate, $A_{ii} = \|b_i\|^2 > 0$. If $r \ge 2$, the random variable $\langle b_i, V\rangle$ has a density that is strictly positive on the interior of $[-\|b_i\|, \|b_i\|]$. Continuity of R_i therefore implies $R_i(s) = 0$ on a neighborhood of the origin. If $r = 1$, then $V \in \{-1, 1\}$ and varying t directly gives the same conclusion, because R_i is even. Thus, for every active coordinate,

$$M_i(s)M_i(-s) = e^{\sigma_i^2 s^2}$$

near the origin. If $\sigma_i^2 = 0$, then $X_i = 0$ almost surely. If $\sigma_i^2 > 0$, an independent copy X_i' gives that $X_i - X_i'$ has the MGF of $\mathcal{N}(0, 2\sigma_i^2)$ near zero and hence has that Gaussian law. The Lévy–Cramér decomposition theorem applied to the independent sum $X_i + (-X_i')$ then implies that X_i is Gaussian.

□

Remark 2.1 (Scope of reflected dominance). Infinite divisibility is sufficient for the dominance assumption, but it is not necessary. Let X be centered, with variance σ^2, and suppose its MGF is finite near zero. If X is infinitely divisible with Lévy measure ν, the Lévy–Khintchine formula gives

$$\log M_X(s) + \log M_X(-s) - \sigma^2 s^2 = 2\int_{\mathbb{R}} \left(\cosh(sx) - 1 - \frac{s^2x^2}{2}\right)\nu(dx) \ge 0$$

where the MGFs are finite. The linear compensation terms cancel, and the Gaussian part cancels against its contribution to the variance. See Sato [21] for the Lévy–Khintchine representation.

For a bounded example outside the infinitely divisible class, take $X = 3$ with probability $1/10$ and $X = -1/3$ with probability $9/10$. Then $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, and $\mathbb{E}X^4 - 3 = 46/9$. Its reflected defect satisfies

$$R_X(s) = \frac{23}{54}s^4 + O(s^6),$$

so $R_X(s) \ge 0$ on a neighborhood of zero. A nondegenerate bounded law cannot be infinitely divisible: if its support has diameter d and it is the sum of n independent identically distributed variables, each summand has support diameter d/n. Its variance is at most $d^2/(4n^2)$, so the variance of the sum is at most $d^2/(4n)$ for every n, a contradiction. Here the variance bound follows by measuring the summand from the midpoint of its support.

Dominance is not a consequence of symmetry either. For a symmetric Rademacher variable,

$$R_X(s) = 2 \log \cosh s - s^2 = -\frac{s^4}{6} + O(s^6) < 0$$

for all sufficiently small nonzero s. Thus the assumption is distinct from the symmetry conditions in the classical sample-variance results.

3 Reflection stability

Throughout this section let X' be an independent copy of X, set

$$Y = X - X', \qquad G \sim \mathcal{N}(0, 2),$$

and define

$$F(z) := M_Y(z) - e^{z^2}, \qquad L := \log(1/\Delta).$$

For probability measures μ and ν, write

$$d_{\mathrm{TV}}(\mu, \nu) := \sup_A |\mu(A) - \nu(A)|.$$

All constants in this section may depend on the fixed radius ρ, but not on Δ.

The proof has four stages. Local control of M_Y first gives Gaussian moment comparison through order $2m$, where $m \asymp L/\log L$. These moments control the tail of X beyond $N = B\sqrt{m}$ and approximate the reflected characteristic function on an interval of length comparable to $\sqrt{m}$. After truncation, a second propagation argument produces a zero-free disk of radius $R = a_0\sqrt{m}$ for one factor. Coefficient bounds for its analytic logarithm then give a Kolmogorov error of order R^{-1}. The balance $m \log m \asymp L$ is the source of the modulus in Theorem 1.1. The constants B and a_0 are fixed independently of Δ in Appendix A.

3.1 Automatic exponential integrability

Lemma 3.1 (Local reflected control supplies an exponential envelope) *Under the hypotheses of Theorem 1.1, if $\Delta \le 1$, then*

$$M_X(s) \le e^{s^2 + \Delta} \qquad (|s| \le \rho)$$

and

$$\mathbb{E}e^{\rho|X|} \le M_X(\rho) + M_X(-\rho) \le 2e^{\rho^2+1}. \tag{7}$$

Consequently $\mathbb{E}e^{\rho|Y|}$ is bounded by a constant depending only on ρ.

Proof Since $\mathbb{E}X = 0$, Jensen's inequality gives $M_X(s) \geq e^{s\mathbb{E}X} = 1$ for every s in the domain of the MGF. Hence $\log M_X(s) \geq 0$. The defect bound gives

$$\log M_X(s) + \log M_X(-s) \leq s^2 + \Delta,$$

so each of the two nonnegative logarithms is at most $s^2 + \Delta$. This proves the first bound. Moreover $e^{\rho|x|} \leq e^{\rho x} + e^{-\rho x}$, which gives (7). Finally,

$$\mathbb{E}e^{\rho|Y|} \leq \mathbb{E}e^{\rho(|X|+|X'|)} = (\mathbb{E}e^{\rho|X|})^2.$$

$\square$

3.2 Half-disk propagation

Lemma 3.2 (Two-constants estimate on a half-disk) *Let $D_R^+ = \{z \in \mathbb{C} : |z| < R,\ \Im z > 0\}$. Suppose that H is holomorphic on a neighborhood of $\overline{D_R^+}$ and satisfies*

$$|H(x)| \leq a \quad (|x| \leq R), \qquad |H(z)| \leq b \quad (|z| = R,\ \Im z \geq 0),$$

where $0 < a \leq b$. Then for every fixed $0 < \eta < 1$ there is $\omega_\eta > 0$, depending only on η, such that

$$\log|H(z)| \leq \omega_\eta \log a + (1 - \omega_\eta)\log b, \qquad |z| \leq \eta R, \quad \Im z \geq 0.$$

The same estimate holds in the lower half-disk.

Proof For $\delta > 0$, the function $u_\delta(z) := \log(|H(z)| + \delta)$ is subharmonic. Its boundary values are at most $\log(a + \delta)$ on the diameter and at most $\log(b + \delta)$ on the semicircle. The two-constants theorem for subharmonic functions bounds u_δ by the corresponding harmonic-measure combination; see Ransford [22]. Under the scaling $z = Rw$, the half-disk D_R^+ becomes the unit half-disk, and the inner region $|z| \leq \eta R$ becomes $|w| \leq \eta$. The harmonic measure of the diameter is positive on this compact inner region, so its minimum is a constant $\omega_\eta > 0$ depending only on η, not on R, m, or Δ. Letting $\delta \downarrow 0$ gives the assertion. $\square$

3.3 High-order moments of the reflected variable

Lemma 3.3 (Local defect to high-order moments) *For every $A > 0$ there is $c_0 = c_0(A, \rho) > 0$ such that, with*

$$m = \left\lfloor c_0 \frac{L}{\log L} \right\rfloor,$$

all sufficiently small Δ satisfy

$$\left|\mathbb{E}Y^k - \mathbb{E}G^k\right| \leq e^{-Am\log m}, \qquad 0 \leq k \leq 2m.$$

In particular,

$$\mathbb{E}|Y|^{2m} \leq (Cm)^m.$$

Proof Set $\rho_0 = \rho/2$ and put $D_{\rho_0}^\pm := \{z : |z| < \rho_0,\ \pm\Im z > 0\}$. By Lemma 3.1, M_Y is holomorphic in the strip $|\Re z| < \rho$, and on the two semicircular arcs $|z| = \rho_0$,

$$|M_Y(z)| \leq \mathbb{E}e^{\rho_0|Y|} \leq C_\rho.$$

Thus $|F(z)| \leq B_0(\rho)$ there. On the real diameter, the defect hypothesis gives

$$|F(s)| = e^{s^2}|e^{R_X(s)} - 1| \leq C_0(\rho)\Delta, \qquad |s| \leq \rho_0.$$

For sufficiently small Δ, the diameter bound is no larger than the arc bound. Apply Lemma 3.2 with $\eta = 1/2$ to $D^+_{\rho_0}$ and $D^-_{\rho_0}$. Set $r_0 := \rho/4$ and $\omega_0 := \omega_{1/2}$. Thus there are constants $r_0 > 0$, $\omega_0 > 0$, and $C_1 < \infty$, depending only on ρ, such that

$$\sup_{|z|\le r_0} |F(z)| \le C_1 \Delta^{\omega_0}.$$

The Cauchy estimate consequently gives

$$|F^{(k)}(0)| \le C_1 k! r_0^{-k} \Delta^{\omega_0}.$$

For $k \le 2m$, use $\log(k!) \le k \log k$ and $k \log k \le 2m \log(2m)$ to obtain

$$\begin{aligned}\log\big(C_1 k! r_0^{-k} \Delta^{\omega_0}\big) &\le -\omega_0 L + 2m \log(2m) + 2m |\log r_0| + O_\rho(1) \\ &\le -\omega_0 L + O_\rho(c_0 L),\end{aligned}$$

when $m = \lfloor c_0 L / \log L \rfloor$ and L is large. Since $m \log m \le c_0 L + o(L)$, choosing c_0 sufficiently small, depending on (A, ρ), gives $|F^{(k)}(0)| \le e^{-Am \log m}$. Since $F^{(k)}(0) = \mathbb{E}Y^k - \mathbb{E}G^k$, the moment conclusion follows. Finally,

$$\mathbb{E}G^{2m} = \frac{(2m)!}{m!} \le (Cm)^m,$$

and the negligible moment mismatch gives the stated $2m$-th moment bound. □

3.4 Tail bound and truncation

Lemma 3.4 (Square-root tail lifting) *There are constants $C, c > 0$ with the following property. For every sufficiently large fixed B, if $N = B\sqrt{m}$, then*

$$\mathbb{P}(|X| > N) \le C e^{-\beta_B m},$$

where $\beta_B \to \infty$ as $B \to \infty$. If X_N denotes the conditional law $X \mid \{|X| \le N\}$, then

$$d_{\mathrm{TV}}(\mathcal{L}(X), \mathcal{L}(X_N)) \le C e^{-\beta_B m}.$$

Moreover

$$|\mathbb{E}X_N| + |\operatorname{Var}(X_N) - 1| \le C(1 + N^2) e^{-cN}.$$

Proof For every $p \ge 1$, conditional Jensen and $\mathbb{E}(Y \mid X) = X$ give

$$|X|^p = |\mathbb{E}(Y \mid X)|^p \le \mathbb{E}(|Y|^p \mid X),$$

and hence $\mathbb{E}|X|^p \le \mathbb{E}|Y|^p$. Taking $p = 2m$ and using Lemma 3.3 gives $\|X\|_{2m} \le C\sqrt{m}$. Therefore, for $N = B\sqrt{m}$,

$$\mathbb{P}(|X| > N) \le \left(\frac{\|X\|_{2m}}{N}\right)^{2m} \le C \left(\frac{C_0}{B^2}\right)^m.$$

Thus one may take $\beta_B = \frac{1}{2}\log(B^2/C_0)$ for sufficiently large B. If $p_N = \mathbb{P}(|X| > N)$, conditioning gives

$$d_{\mathrm{TV}}(\mathcal{L}(X), \mathcal{L}(X_N)) \le p_N.$$

By (7), the exponential envelope implied by the defect bound also gives, for some $c = c(\rho) > 0$,

$$\mathbb{E}\big(|X| \mathbf{1}_{\{|X|>N\}}\big) \le C e^{-cN}, \qquad \mathbb{E}\big(X^2 \mathbf{1}_{\{|X|>N\}}\big) \le C(1 + N^2) e^{-cN}.$$

Since $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$,

$$|\mathbb{E}X_N| \le \frac{Ce^{-cN}}{1 - p_N}, \qquad |\mathbb{E}X_N^2 - 1| \le \frac{p_N + C(1 + N^2)e^{-cN}}{1 - p_N}.$$

It follows that $|\mathbb{E}X_N| + |\operatorname{Var}(X_N) - 1| \le C(1 + N^2)e^{-cN}$ after increasing C. □

3.5 Product approximation on a growing interval

Lemma 3.5 (Real-axis product approximation) *For B fixed sufficiently large, there is $a_0 = a_0(B, \rho) > 0$ such that, with*

$$R = a_0\sqrt{m}$$

and φ_N the characteristic function of X_N,

$$\sup_{|t|\le 2R} \left|\varphi_N(t)\varphi_N(-t) - e^{-t^2}\right| \le e^{-\Gamma m}$$

for the finite target exponent $\Gamma > 4$ selected in Appendix A.

Proof Use the $2m-1$ Taylor polynomial of the characteristic function of $Y = X - X'$ and the real-variable remainder

$$\left|e^{iu} - \sum_{k=0}^{2m-1} \frac{(iu)^k}{k!}\right| \le \frac{|u|^{2m}}{(2m)!}.$$

Taylor's inequality and the moment comparison give

$$\begin{aligned}|\varphi_Y(t) - e^{-t^2}| \le{}& \frac{|t|^{2m}\mathbb{E}|Y|^{2m}}{(2m)!} + \frac{|t|^{2m}\mathbb{E}|G|^{2m}}{(2m)!} \\ &+ \sum_{k=0}^{2m-1} \frac{|t|^k}{k!}|\mathbb{E}(iY)^k - \mathbb{E}(iG)^k|,\end{aligned}$$

where $G \sim \mathcal{N}(0, 2)$. On $|t| \le 2R$, each Taylor remainder is at most $(C_T a_0^2)^m$ by Stirling's formula. The moment-mismatch term is at most

$$e^{-Am\log m} \sum_{k=0}^{2m-1} \frac{|t|^k}{k!} \le \exp\{-Am\log m + 2a_0\sqrt{m}\}.$$

Choose B so that $\beta_B > 4\Gamma$ and then choose a_0 so that $C_T a_0^2 < e^{-4\Gamma}$. These choices are compatible with the complex propagation step, as recorded in Appendix A.

For completeness, write $p_N = \mathbb{P}(|X| > N)$. Under our convention for total variation, integration of a complex-valued function bounded by one gives

$$|\varphi_X(t) - \varphi_N(t)| \le 2p_N.$$

Since characteristic functions have modulus at most one on the real axis,

$$|\varphi_X(t)\varphi_X(-t) - \varphi_N(t)\varphi_N(-t)| \le 4p_N \le 4Ce^{-\beta_B m}.$$

Combining this with $\varphi_Y(t) = \varphi_X(t)\varphi_X(-t)$ proves, uniformly on $|t| \le 2R$,

$$|\varphi_N(t)\varphi_N(-t) - e^{-t^2}| \le 2e^{-4\Gamma m} + e^{-Am\log m + 2a_0\sqrt{m}} + 4Ce^{-\beta_B m} \le e^{-\Gamma m}$$

for all sufficiently large m. □

3.6 A zero-free disk

Lemma 3.6 (Growing zero-free disk) *The constants B and a_0 can be chosen so that, for $R = a_0\sqrt{m}$,*

$$\varphi_N(z) \ne 0, \qquad |z| \le R.$$

More precisely, for some $\gamma_* > a_0^2$,

$$\sup_{|z|\le R} \left|\varphi_N(z)\varphi_N(-z) - e^{-z^2}\right| \le e^{-\gamma_* m}.$$

Proof Since $|X_N| \le N$, φ_N is entire and

$$|\varphi_N(z)| \le e^{N|\Im z|}.$$

Hence, for

$$\mathcal{F}_N(z) := \varphi_N(z)\varphi_N(-z) - e^{-z^2},$$

the explicit growth bound

$$\log|\mathcal{F}_N(z)| \le \log 2 + 4NR + 4R^2 \le \log 2 + 4(Ba_0 + a_0^2)m, \qquad |z| \le 2R,$$

holds. On the real diameter, Lemma 3.5 gives $\log|\mathcal{F}_N| \le -\Gamma m$. Apply Lemma 3.2 with radius $2R$ and $\eta = 1/2$ in the upper and lower half-disks. With $\omega_0 := \omega_{1/2}$, this gives

$$\log|\mathcal{F}_N(z)| \le -\gamma_{\rm raw} m + O(1), \qquad \gamma_{\rm raw} := \omega_0\Gamma - 4(1-\omega_0)(Ba_0 + a_0^2).$$

Here B has already been fixed to control the truncation error. We now decrease a_0, keeping B fixed, so that $\gamma_{\rm raw} > a_0^2$. Decreasing a_0 also improves the Taylor estimate in Lemma 3.5. Appendix A records these choices explicitly. Choose once and for all

$$a_0^2 < \gamma_* < \gamma_{\rm raw}.$$

For all sufficiently large m, the $O(1)$ term is then absorbed and

$$|\mathcal{F}_N(z)| \le e^{-\gamma_* m}, \qquad |z| \le R.$$

Since

$$\inf_{|z|\le R} |e^{-z^2}| \ge e^{-R^2} = e^{-a_0^2 m},$$

we have

$$\frac{|\mathcal{F}_N(z)|}{|e^{-z^2}|} \le e^{-(\gamma_* - a_0^2)m} < \frac{1}{2}$$

for all sufficiently large m. Hence

$$|\varphi_N(z)\varphi_N(-z)| \ge \frac{1}{2}|e^{-z^2}| > 0,$$

and therefore $\varphi_N(z) \ne 0$ on the disk. $\square$

3.7 From zero-freeness to Gaussian approximation

Lemma 3.7 (From zero-freeness to Gaussian approximation) *Under the conclusions of Lemma 3.6,*

$$d_{\rm K}(X_N, \mathcal{N}(\mu_N, \sigma_N^2)) \le \frac{C}{R},$$

where $\mu_N = \mathbb{E}X_N$ *and* $\sigma_N^2 = \mathrm{Var}(X_N)$*. Consequently*

$$d_{\rm K}(X, \mathcal{N}(0,1)) \le \frac{C}{\sqrt{m}}.$$

Proof The disk $D_R := \{z : |z| < R\}$ is simply connected, and Lemma 3.6 gives $\varphi_N \ne 0$ there. Hence there is a unique analytic logarithm on D_R satisfying $\psi_N(0) = 0$; write

$$\psi_N(z) = \log\varphi_N(z), \qquad \psi_N(0) = 0.$$

The bounded support gives $\Re\psi_N(z) = \log|\varphi_N(z)| \le NR$. Set

$$H(z) := NR - \psi_N(z), \qquad \Re H(z) \ge 0.$$

For $\psi_N(z) = \sum_{k\geq 1} a_k z^k$, define

$$P(w) := \frac{H(Rw)}{NR} = 1 - \sum_{k\geq 1} \frac{a_k R^{k-1}}{N} w^k, \qquad |w| < 1.$$

Then $P(0) = 1$ and $\Re P \geq 0$. The Carathéodory coefficient theorem bounds every nonconstant coefficient of P by 2 in modulus; see Duren [23]. Hence

$$|a_k| \leq 2NR^{1-k}, \qquad k \geq 1.$$

This estimate retains the support scale N, which is comparable to R in the present argument. Since $a_1 = i\mu_N$ and $a_2 = -\sigma_N^2/2$, for every fixed $0 < \theta < 1$,

$$\left|\psi_N(t) - i\mu_N t + \frac{\sigma_N^2 t^2}{2}\right| \leq \sum_{k\geq 3} 2NR^{1-k}|t|^k \leq \frac{2N|t|^3}{R^2(1-|t|/R)} \leq C\frac{|t|^3}{R}, \qquad |t| \leq \theta R,$$

because $N/R = B/a_0$ is fixed. Define the cubic remainder

$$E_N(t) := \psi_N(t) - i\mu_N t + \frac{\sigma_N^2 t^2}{2}.$$

By Lemma 3.4, for all sufficiently large m,

$$|\mu_N| = o(R^{-1}), \qquad \frac{1}{2} \leq \sigma_N^2 \leq \frac{3}{2}, \qquad |E_N(t)| \leq C\frac{|t|^3}{R}.$$

Choose θ so small that $C\theta \leq 1/8$. Then, on $|t| \leq \theta R$,

$$\Re\psi_N(t) = -\frac{\sigma_N^2 t^2}{2} + \Re E_N(t) \leq -\frac{t^2}{4} + \frac{t^2}{8} = -\frac{t^2}{8}.$$

Writing

$$\varphi_N(t) = e^{i\mu_N t - \sigma_N^2 t^2/2} e^{E_N(t)},$$

and using $|e^z - 1| \leq |z|e^{|z|}$ gives Gaussian damping in the difference:

$$\left|\varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2}\right| \leq C\frac{|t|^3}{R} e^{-ct^2}.$$

Let g_N be the density of $\mathcal{N}(\mu_N, \sigma_N^2)$. The Esseen smoothing inequality in the present normalization (Esseen [24]) is

$$\begin{aligned} d_{\mathrm{K}}(X_N, \mathcal{N}(\mu_N, \sigma_N^2)) &\leq \frac{1}{\pi}\int_{-T}^{T} \left|\frac{\varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2}}{t}\right| dt + \frac{24\|g_N\|_\infty}{\pi T} \\ &= \frac{1}{\pi}\int_{-T}^{T} \left|\frac{\varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2}}{t}\right| dt + O(T^{-1}), \end{aligned}$$

where $\|g_N\|_\infty = (\sqrt{2\pi}\sigma_N)^{-1}$ and the last form uses $\sigma_N^2 \geq 1/2$. With $T = \theta R$, the integral is bounded by

$$\frac{C}{R}\int_{-T}^{T} t^2 e^{-ct^2}\, dt \leq \frac{C}{R}\int_{\mathbb{R}} t^2 e^{-ct^2}\, dt = O(R^{-1}),$$

and the cutoff term is also $O(R^{-1})$.

It remains to compare the nearby Gaussian with the standard one. For σ in a fixed compact subset of $(0, \infty)$,

$$d_{\mathrm{K}}(\mathcal{N}(\mu, \sigma^2), \mathcal{N}(0,1)) \leq C\big(|\mu| + |\sigma - 1|\big). \tag{8}$$

Indeed, the translation term is bounded by $C|\mu|$ using the bounded Gaussian density, while the mean-value theorem in σ gives

$$\sup_x |\Phi(x/\sigma) - \Phi(x)| \le C|\sigma - 1|,$$

because $\sup_y |y|\phi(y) < \infty$. Since $\sigma_N^2 \in [1/2, 3/2]$, also $|\sigma_N - 1| \le C|\sigma_N^2 - 1|$. Thus Lemma 3.4 and (8) show that the Gaussian parameter shift is $o(R^{-1})$. Finally,

$$d_{\mathrm{K}}(X, X_N) \le d_{\mathrm{TV}}(\mathcal{L}(X), \mathcal{L}(X_N)) = o(R^{-1}),$$

so the triangle inequality yields the consequence for X. $\square$

Proof of Theorem 1.1 The exponential envelope follows from Lemma 3.1. Apply Lemmas 3.3 to 3.7. Since

$$m \asymp_\rho \frac{L}{\log L}, \qquad L = \log(1/\Delta),$$

we have

$$\frac{1}{\sqrt{m}} \asymp_\rho \sqrt{\frac{\log L}{L}},$$

which is the claimed rate. $\square$

4 Optimality and obstruction results

4.1 A symmetric Laguerre lower-bound family

We prove sharpness within a symmetric class of densities uniformly comparable with the Gaussian density. The perturbation cancels the first $2m - 1$ moments, giving a reflected-MGF defect of factorial size on each fixed interval, while its distribution function differs from the Gaussian one by order $m^{-1/2}$. We use the following Laguerre bound, proved in Appendix B from the estimates of Imekraz, Robert and Thomann [25, Proposition 3.2].

Lemma 4.1 (Uniform even Laguerre envelope)

$$C_E := \sup_{m \ge 2,\, x \in \mathbb{R}} \sqrt{m}\, e^{-x^2} \left| L_m^{(-1/2)}(3x^2/2) \right| < \infty. \tag{9}$$

Proof See Appendix B. $\square$

Fix $0 < \epsilon < 1/4$, let ϕ be the standard Gaussian density, and define, for $m \ge 2$,

$$h_m(x) := \frac{(-1)^m \sqrt{m}}{C_E} e^{-x^2} L_m^{(-1/2)}(3x^2/2), \qquad f_m(x) := \phi(x)(1 + \epsilon h_m(x)).$$

By (9), $|h_m| \le 1$, so

$$(1 - \epsilon)\phi(x) \le f_m(x) \le (1 + \epsilon)\phi(x). \tag{10}$$

In particular f_m is nonnegative and has exponential moments of every fixed order uniformly in m.

The perturbation is even. With $r = 3x^2/2$, Laguerre orthogonality for parameter $-1/2$ (Szegő [26, Chapter V]) gives, for $0 \le k < m$,

$$\int_{\mathbb{R}} x^{2k} h_m(x)\phi(x)\,dx = C_{m,k} \int_0^\infty r^{k-1/2} e^{-r} L_m^{(-1/2)}(r)\,dr = 0,$$

where $C_{m,k} \neq 0$ is an irrelevant deterministic factor. The cases $k = 0$ and $k = 1$ show that f_m has total mass one and variance one, while evenness gives mean zero. Let W_m have density f_m.

To compute the MGF exactly, set

$$J_m(s) := \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} e^{sx - 3x^2/2} L_m^{(-1/2)}(3x^2/2)\,dx.$$

The Laguerre generating function, first integrated for $|t|$ sufficiently small and then continued analytically to $|t| < 1$, gives

$$\begin{aligned}
\sum_{m \ge 0} J_m(s) t^m &= \frac{(1-t)^{-1/2}}{\sqrt{2\pi}} \int_{\mathbb{R}} \exp\left(sx - \frac{3x^2}{2(1-t)}\right) dx \\
&= \frac{1}{\sqrt{3}} \exp\left(\frac{s^2(1-t)}{6}\right) \\
&= \frac{1}{\sqrt{3}} e^{s^2/6} \sum_{m \ge 0} \frac{(-s^2/6)^m}{m!} t^m.
\end{aligned}$$

Thus

$$J_m(s) = \frac{1}{\sqrt{3}} e^{s^2/6} \frac{(-s^2/6)^m}{m!}.$$

After multiplying by the factor $(-1)^m \sqrt{m}/C_E$ in h_m and dividing by the Gaussian MGF $e^{s^2/2}$, we obtain

$$M_{W_m}(s) = e^{s^2/2}(1 + v_m(s)), \qquad v_m(s) := \frac{\epsilon\sqrt{m}}{\sqrt{3}\,C_E} e^{-s^2/3} \frac{(s^2/6)^m}{m!}. \tag{11}$$

In particular $v_m(s) \ge 0$ on the real axis. Since W_m is symmetric,

$$R_{W_m}(s) = 2\log(1 + v_m(s)) \ge 0. \tag{12}$$

Fix any $\rho > 0$. For $m > \rho^2/3$, the function $s \mapsto e^{-s^2/3} s^{2m}$ is increasing on $[0, \rho]$, since its logarithmic derivative is $2m/s - 2s/3 > 0$ for $0 < s \le \rho$. Therefore

$$\Delta_m := \Delta_\rho(W_m) = 2\log(1 + v_m(\rho)).$$

Here

$$v_m(\rho) = \frac{\epsilon\sqrt{m}}{\sqrt{3}\,C_E} e^{-\rho^2/3} \frac{(\rho^2/6)^m}{m!} \longrightarrow 0.$$

Thus $\Delta_m \asymp v_m(\rho)$, and Stirling's formula yields

$$\log \frac{1}{\Delta_m} = m \log m + O_\rho(m). \tag{13}$$

Hence

$$\sqrt{\frac{\log\log(1/\Delta_m)}{\log(1/\Delta_m)}} \asymp m^{-1/2}.$$

It remains to prove a Kolmogorov discrepancy of this order. The value at the origin is

$$L_m^{(-1/2)}(0) = \frac{(1/2)_m}{m!} = \frac{1}{4^m}\binom{2m}{m} \asymp m^{-1/2}.$$

Moreover,

$$\frac{L_m^{(-1/2)}(r)}{L_m^{(-1/2)}(0)} = \sum_{k=0}^{m} \frac{(-m)_k}{(1/2)_k\, k!} r^k. \tag{14}$$

Choose a fixed $c_0 > 0$ so small that $e^{2c_0} - 1 \leq 1/2$. For $0 \leq r \leq c_0/m$, using $|(-m)_k| \leq m^k$ and $(1/2)_k \geq 2^{-k} k!$ gives

$$\left| \frac{L_m^{(-1/2)}(r)}{L_m^{(-1/2)}(0)} - 1 \right| \leq \sum_{k\geq 1} \frac{(2mr)^k}{(k!)^2} \leq e^{2c_0} - 1 \leq \frac{1}{2}.$$

Therefore $L_m^{(-1/2)}(r)$ keeps the sign of its value at the origin and is at least half as large there. Put

$$x_m := \sqrt{\frac{2c_0}{3m}}.$$

For $0 \leq x \leq x_m$, (9) and the preceding lower bound show that $h_m(x)$ has the fixed sign $(-1)^m$ and

$$|h_m(x)| \geq c_1$$

with $c_1 > 0$ independent of m. Since $f_m - \phi = \epsilon h_m \phi$ is even and has total integral zero, its integral over $(-\infty, 0]$ vanishes. Consequently

$$\begin{aligned} d_{\mathrm{K}}(W_m, \mathcal{N}(0,1)) &\geq |F_{W_m}(x_m) - \Phi(x_m)| \\ &= \epsilon \left| \int_0^{x_m} h_m(x)\phi(x)\,dx \right| \geq c_2 x_m \geq c_3 m^{-1/2}. \end{aligned}$$

Together with (13), this proves Theorem 1.2. In particular no uniform power-law modulus $C\Delta^\alpha$, $\alpha > 0$, can replace the logarithmic modulus in Theorem 1.1.

Since $M_{W_m}(-s) = M_{W_m}(s) > 0$ for real s, the lower bound persists even when the MGF factor is the positive square root of its reflected product. It is therefore an obstruction to recovery from local transform data, not only an effect of lost Fourier phase.

4.2 Failure of total-variation stability

Throughout this subsection, $d_{\mathrm{TV}}(\mu,\nu) := \sup_A |\mu(A) - \nu(A)|$.

Proposition 4.2 (Strong-distance obstruction) *There is no uniform total-variation stability theorem under the assumptions of Theorem 1.1: a standardized Poisson lattice family has reflected defect tending to zero on every fixed interval while its total-variation distance from every absolutely continuous Gaussian law equals one.*

Proof Let $N_\lambda \sim \mathrm{Poisson}(\lambda)$, with $\lambda \geq 1$, and $X_\lambda = (N_\lambda - \lambda)/\sqrt{\lambda}$. Its centered cumulant generating function is

$$K_\lambda(s) = \lambda \left(e^{s/\sqrt{\lambda}} - 1 - \frac{s}{\sqrt{\lambda}} \right).$$

Consequently, uniformly on every fixed compact s-interval,

$$\begin{aligned} R_{X_\lambda}(s) &= K_\lambda(s) + K_\lambda(-s) - s^2 \\ &= 2\lambda \left(\cosh(s/\sqrt{\lambda}) - 1 - \frac{s^2}{2\lambda} \right) = \frac{s^4}{12\lambda} + O_\rho(\lambda^{-2}). \end{aligned}$$

For fixed $b > 0$, the inequality $e^u - 1 - u \leq u^2 e^{|u|}/2$ gives

$$\mathbb{E}e^{b|X_\lambda|} \leq \mathbb{E}e^{bX_\lambda} + \mathbb{E}e^{-bX_\lambda} \leq 2\exp\left(\frac{b^2}{2} e^{b/\sqrt{\lambda}} \right),$$

so the family even has a fixed exponential-integrability bound for each fixed b. Nevertheless X_λ is lattice-valued, whereas $\mathcal{N}(0,1)$ is absolutely continuous, so

$$d_{\mathrm{TV}}(\mathcal{L}(X_\lambda), \mathcal{N}(0,1)) = 1.$$

$\square$

This lattice example separates the metric issue from the sharp Kolmogorov modulus. Related failures of strong-distance stability for Gaussian convolutions, including absolutely continuous symmetric examples, were established by Bobkov, Chistyakov and Götze [5]. Their regularization results [27] show how additional Gaussian smoothing restores total-variation and entropic stability in the convolution setting.

5 Sample variance

Let $X_1, \ldots, X_n$ be i.i.d. copies of a centered variance-one random variable X, let $\bar{X} = n^{-1} \sum_i X_i$, and set

$$Q_n = \sum_{i=1}^{n} (X_i - \bar{X})^2.$$

This section applies reflection stability to Q_n. We first record the exact consequence of Theorem 1.3. For stability, spherical averaging provides an intermediate transform quantity, which will be bounded in terms of the usual Kolmogorov distance in Section 5.3.

5.1 The exact consequence

The exact sample-variance consequence of the quadratic-form theorem is the following.

Corollary 5.1 (Exact sample-variance characterization under dominance) *Let $n \geq 2$ and let $X_1, \ldots, X_n$ be i.i.d., centered, and variance one. If*

$$M_X(s)M_X(-s) \geq e^{s^2}$$

for all s in some neighborhood of the origin, and

$$\sum_{i=1}^{n}(X_i - \bar{X})^2 \sim \chi^2_{n-1},$$

then $X_i \sim \mathcal{N}(0, 1)$.

Proof Take $A = I_n - n^{-1}\mathbf{1}\mathbf{1}^\mathsf{T}$ in Theorem 1.3. This is a rank-$(n-1)$ orthogonal projection with every diagonal entry equal to $(n-1)/n > 0$, and $X^\mathsf{T}AX = Q_n$. $\square$

As in Theorem 1.3, the chi-square law implies MGF existence for every coordinate. When $n = 2$, dominance is not needed for the exact conclusion: $Q_2 = (X_1 - X_2)^2/2$, and the symmetry of $X_1 - X_2$ identifies its law as $\mathcal{N}(0, 2)$. Cramér's theorem then gives Gaussianity of X.

5.2 From spherical discrepancy to reflected defect

For $r \geq 1$, let V be uniform on S^{r-1}. Recall the spherical transform

$$\Psi_r(z) := \mathbb{E}_V e^{zV_1},$$

and define, for a nonnegative random variable Q with finite displayed expectations,

$$\mathfrak{D}_{r,\tau}(Q) := \sup_{|t|\leq\tau} \left| e^{-t^2/2}\mathbb{E}\Psi_r(t\sqrt{Q}) - 1 \right|.$$

This quantity vanishes for $Q \sim \chi^2_r$. It is a transform discrepancy, not a probability metric. Its role is to connect the quadratic form to an average of the one-dimensional reflected defect. The next subsection bounds it by Kolmogorov distance under an exponential envelope.

The quantitative statement based on the spherical observable is the next theorem.

Theorem 5.2 (Sample-variance stability under spherical Laplace discrepancy) *Fix $n \geq 3$, $\tau, K > 0$, and $0 < \rho' < \tau\sqrt{(n-1)/n}$. There are constants $C < \infty$ and $\varepsilon_0 \in (0, e^{-e})$, depending only on (n, τ, K, ρ'), such that the following holds. Let $X_1, \ldots, X_n$ be i.i.d. copies of a centered variance-one X with $\mathbb{E}e^{2\tau|X|} \leq K$, and assume $R_X(s) \geq 0$ for $|s| \leq \tau$. If*

$$Q_n = \sum_{i=1}^{n}(X_i - \bar{X})^2, \qquad \mathfrak{D}_{n-1,\tau}(Q_n) \leq \varepsilon, \qquad 0 < \varepsilon \leq \varepsilon_0,$$

then

$$d_{\mathrm{K}}(X, \mathcal{N}(0,1)) \le C\sqrt{\frac{\log\log(1/\varepsilon)}{\log(1/\varepsilon)}}.$$

Proof Let $r = n-1$, let $P = I_n - n^{-1}\mathbf{1}\mathbf{1}^{\mathsf{T}}$, and choose $B \in \mathbb{R}^{r\times n}$ with $B^{\mathsf{T}}B = P$ and $BB^{\mathsf{T}} = I_r$. If V is uniform on S^{r-1}, then every column b_i of B has norm

$$q := \|b_i\| = \sqrt{\frac{n-1}{n}},$$

and $\langle b_i, V\rangle$ has the same distribution as qV_1. Put $a = \tau q$ and define

$$D_X(s) := \frac{1}{2}R_X(s) = \frac{1}{2}\log\frac{M_X(s)M_X(-s)}{e^{s^2}}.$$

Under dominance, $D_X \ge 0$ on $[-\tau, \tau]$.

The exact spherical transform gives, for $|t| \le \tau$,

$$e^{-t^2/2}\mathbb{E}\Psi_r(t\sqrt{Q_n}) = \mathbb{E}_V \exp\left(\sum_{i=1}^n \left[\log M_X(t\langle b_i, V\rangle) - \frac{t^2\langle b_i, V\rangle^2}{2}\right]\right). \tag{15}$$

Let $H_t(V)$ denote the exponent on the right. The identity uses $BB^{\mathsf{T}} = I_r$, which gives $\sum_i \langle b_i, V\rangle^2 = 1$. Each $\langle b_i, V\rangle$ has the law of qV_1, and symmetry gives

$$\mathbb{E}_V H_\tau(V) = n\mathbb{E}D_X(aV_1) \ge 0.$$

Jensen's inequality and the definition of the discrepancy now give

$$1 \le \exp\{n\mathbb{E}D_X(aV_1)\} \le e^{-\tau^2/2}\mathbb{E}\Psi_r(\tau\sqrt{Q_n}) \le 1 + \mathfrak{D}_{n-1,\tau}(Q_n).$$

Taking logarithms yields

$$n\mathbb{E}D_X(aV_1) \le \log\bigl(1 + \mathfrak{D}_{n-1,\tau}(Q_n)\bigr). \tag{16}$$

We next turn this averaged estimate into a bound on an interior interval.

For the pointwise transfer, the density of V_1 is

$$f_n(v) = \frac{\Gamma((n-1)/2)}{\sqrt{\pi}\,\Gamma((n-2)/2)}(1-v^2)^{(n-4)/2}, \qquad -1 < v < 1.$$

Fix $0 < \rho' < a$ and choose $0 < h < a - \rho'$. On $[-(\rho'+h), \rho'+h]$ the scaled density of aV_1 is bounded below by

$$m_h := \frac{1}{a}\frac{\Gamma((n-1)/2)}{\sqrt{\pi}\,\Gamma((n-2)/2)}\left(1 - \left(\frac{\rho'+h}{a}\right)^2\right)^{\max(n-4,0)/2}.$$

The exponential envelope implies that D_X is Lipschitz there. One may take

$$L_h := \frac{K}{e\tau} + \rho' + h,$$

because $M_X(s) \ge 1$ and $|x|e^{\tau|x|} \le (e\tau)^{-1}e^{2\tau|x|}$. If $M_* := \sup_{|s|\le\rho'} D_X(s)$, choose $s_0 \in [-\rho', \rho']$ with $D_X(s_0) = M_*$. The function D_X is even, nonnegative, and satisfies $D_X(0) = 0$. Since it is L_h-Lipschitz on the enlarged interval,

$$D_X(s) \ge \bigl(M_* - L_h|s - s_0|\bigr)_+.$$

If $M_* = 0$, the desired pointwise bound is immediate. Otherwise $|s_0| \geq M_*/L_h$ because $D_X(0) = 0$. Put $\ell = \min\{h, M_*/L_h\}$. The intervals

$$[s_0 - \ell, s_0 + \ell], \qquad [-s_0 - \ell, -s_0 + \ell]$$

lie in the enlarged interval and have disjoint interiors, since $\ell \leq |s_0|$. On each, integrate the corresponding tent. If $M_* \leq L_h h$, its positive part has half-width M_*/L_h; if $M_* > L_h h$, restrict the tent to half-width h. Integrating against the density lower bound gives

$$2m_h I_{L_h,h}(M_*) \leq \frac{1}{n} \log\bigl(1 + \mathfrak{D}_{n-1,\tau}(Q_n)\bigr),$$

where

$$I_{L,h}(M) := \begin{cases} M^2/L, & M \leq Lh, \\ 2hM - Lh^2, & M > Lh. \end{cases}$$

In both cases $I_{L,h}(M) = \int_{-h}^{h} (M - L|u|)_+ \, du$. Writing $G(y; L, h) = \sqrt{Ly}$ when $y \leq Lh^2$ and $G(y; L, h) = y/(2h) + Lh/2$ otherwise, we obtain

$$\sup_{|s| \leq \rho'} D_X(s) \leq \inf_{0<h<a-\rho'} G\left(\frac{\log(1 + \mathfrak{D}_{n-1,\tau}(Q_n))}{2nm_h}; L_h, h\right). \tag{17}$$

In particular, for fixed n, τ, K, ρ', the right-hand side is $O(\sqrt{\mathfrak{D}_{n-1,\tau}(Q_n)})$ as the discrepancy tends to zero. The strict interior restriction is used by this density-lower-bound argument when $n \geq 5$, because the spherical density vanishes at the endpoints; an endpoint result would require a different local estimate.

Applying (17) to the sample-variance hypothesis gives

$$\Delta_{\rho'}(X) \leq C_1 \sqrt{\varepsilon},$$

for all sufficiently small ε, where $C_1 = C_1(n, \tau, K, \rho')$; here we used $R_X = 2D_X$. Set $\Delta = C_1\sqrt{\varepsilon}$. Then

$$\log(1/\Delta) = \tfrac{1}{2} \log(1/\varepsilon) - \log C_1 \asymp \log(1/\varepsilon), \qquad \log\log(1/\Delta) = \log\log(1/\varepsilon) + O(1).$$

Theorem 1.1, with radius ρ', yields the asserted rate. The exponential envelope was used to obtain the Lipschitz bound that transfers the spherical average to a pointwise defect. ☐

5.3 Transfer from Kolmogorov distance

We now control the intermediate transform discrepancy by Kolmogorov distance. Let $H_r \sim \chi^2_r$.

Lemma 5.3 (Kolmogorov control of the spherical observable) *Fix $r \geq 1$ and $\tau > 0$. Let $Q \geq 0$ and suppose*

$$\mathbb{E} e^{2\tau\sqrt{Q}} \leq A.$$

Then, for

$$\delta := d_{\mathrm{K}}(Q, H_r) \leq 1,$$

one has

$$\mathfrak{D}_{r,\tau}(Q) \leq C_{r,\tau,A} \sqrt{\delta}.$$

Proof If $\delta = 0$, then Q and H_r have the same law and the discrepancy is zero. Suppose $0 < \delta \le 1$. For $|t| \le \tau$, define

$$g_t(q) := e^{-t^2/2}\Psi_r(t\sqrt{q}), \qquad q \ge 0.$$

Because V_1 is symmetric,

$$\Psi_r(t\sqrt{q}) = \mathbb{E}_V \cosh(t\sqrt{q}\,V_1),$$

so g_t is nonnegative and nondecreasing in q. Also

$$0 \le g_t(q) \le e^{|t|\sqrt{q}} \le e^{\tau\sqrt{q}}.$$

If $H_r = \|G\|^2$ with $G \sim \mathcal{N}(0, I_r)$, spherical symmetry gives

$$\mathbb{E}g_t(H_r) = e^{-t^2/2}\mathbb{E}e^{tG_1} = 1.$$

Moreover,

$$\sup_{|t|\le\tau} \mathbb{E}g_t(Q)^2 \le A, \qquad \sup_{|t|\le\tau} \mathbb{E}g_t(H_r)^2 \le \mathbb{E}e^{2\tau\sqrt{H_r}} =: A_{r,\tau} < \infty.$$

For $T \ge 1$, the function $g_t \wedge T$ is bounded and nondecreasing, with total variation at most T. Integration by parts against the difference of the distribution functions therefore gives

$$|\mathbb{E}(g_t(Q) \wedge T) - \mathbb{E}(g_t(H_r) \wedge T)| \le T\delta.$$

The tails satisfy

$$\mathbb{E}(g_t(Q) - T)_+ \le \frac{\mathbb{E}g_t(Q)^2}{T} \le \frac{A}{T},$$

and likewise for H_r. Hence, uniformly in $|t| \le \tau$,

$$|\mathbb{E}g_t(Q) - 1| \le T\delta + \frac{A + A_{r,\tau}}{T}.$$

Taking $T = \delta^{-1/2}$ proves the claim. $\square$

For the sample variance, the exponential envelope already assumed in Theorem 5.2 provides the hypothesis of Lemma 5.3. Indeed, since $Q_n = X^\mathsf{T} P X \le \sum_i X_i^2$,

$$\sqrt{Q_n} \le \left(\sum_i X_i^2\right)^{1/2} \le \sum_i |X_i|,$$

and independence gives

$$\mathbb{E}e^{2\tau\sqrt{Q_n}} \le \prod_{i=1}^{n} \mathbb{E}e^{2\tau|X_i|} \le K^n. \tag{18}$$

We therefore obtain the following standard-distance consequence.

Corollary 5.4 (Sample-variance stability in Kolmogorov distance) *Fix $n \ge 3$, $\tau, K > 0$, and $0 < \rho' < \tau\sqrt{(n-1)/n}$. There exist constants $C < \infty$ and $\delta_0 \in (0, e^{-e})$, depending only on (n, τ, K, ρ'), such that the following holds. Let $X_1, \dots, X_n$ be i.i.d. copies of a centered variance-one X satisfying*

$$\mathbb{E}e^{2\tau|X|} \le K, \qquad R_X(s) \ge 0 \quad (|s| \le \tau).$$

If

$$0 < \delta := d_{\mathrm{K}}(Q_n, \chi^2_{n-1}) \le \delta_0,$$

then

$$d_{\mathrm{K}}(X, \mathcal{N}(0,1)) \le C\sqrt{\frac{\log\log(1/\delta)}{\log(1/\delta)}}.$$

If $\delta = 0$, *then* $X \sim \mathcal{N}(0,1)$.

Proof The case $\delta = 0$ follows from Corollary 5.1. For $\delta > 0$, by (18) and Lemma 5.3,

$$\mathfrak{D}_{n-1,\tau}(Q_n) \le C_0\sqrt{\delta}.$$

Apply Theorem 5.2 with $\varepsilon = C_0\sqrt{\delta}$. Since

$$\log(1/\varepsilon) = \tfrac{1}{2}\log(1/\delta) + O(1), \qquad \log\log(1/\varepsilon) = \log\log(1/\delta) + O(1),$$

the asserted rate follows after changing the constant. □

5.4 Scope and further questions

The application uses dominance to recover pointwise information from a nonnegative spherical average. The reflection theorem itself has no such sign assumption. The restriction $n \ge 3$ in the quantitative argument allows us to use the density of a spherical coordinate; the exact $n = 2$ case was treated separately above.

The sharpness result in Section 4 concerns the reflected-MGF problem and does not establish sharpness of Corollary 5.4. Determining the optimal sample-variance modulus remains a separate question. It would also be useful to weaken the exponential envelope used to control the unbounded spherical transform, or to develop a quantitative counterpart of Theorem 1.3 for non-identically distributed coordinates.

A Parameter selection for the sharp upper bound

This appendix records the order of choices for the proof of Theorem 1.1. Fix $\rho > 0$ and the interior analytic radius $\rho/2$ used in Lemma 3.3. Let $\omega_0 = \omega_{1/2} > 0$ be the harmonic-measure bound from Lemma 3.2; the same value is used in both propagation steps. Choose $A > 8$ and then choose $c_0 = c_0(A, \rho) > 0$ sufficiently small that the Cauchy estimate in Lemma 3.3 dominates all factorial losses; equivalently, after absorbing the fixed ρ-dependent constants, one may require

$$(A+3)c_0 < \omega_0.$$

With $m = \lfloor c_0 L/\log L \rfloor$, this gives the required super-exponential moment accuracy.

Fix a finite target exponent $\Gamma > 4$. Choose the truncation factor B sufficiently large that the tail-lifting estimate has exponent $\beta_B > 4\Gamma$. After B is fixed, let C_T denote the constant in the real-axis Taylor remainder and choose $a_0 > 0$ so small that

$$C_T a_0^2 < e^{-4\Gamma}, \qquad 4(1-\omega_0)(Ba_0 + a_0^2) < \frac{\omega_0 \Gamma}{2}, \qquad a_0^2 < \frac{\omega_0 \Gamma}{4}.$$

Each Taylor remainder is at most $e^{-4\Gamma m}$, and the truncation contribution is at most $4Ce^{-\beta_B m}$. Together with the super-exponential moment mismatch, their sum is at most $e^{-\Gamma m}$ for large m. Thus Lemma 3.5 uses the target exponent Γ, while the complex propagation estimate has raw exponent

$$\gamma_{\text{raw}} := \omega_0 \Gamma - 4(1-\omega_0)(Ba_0 + a_0^2) \geq \frac{\omega_0 \Gamma}{2} > a_0^2.$$

Choose any fixed γ_* with

$$a_0^2 < \gamma_* < \gamma_{\text{raw}}.$$

The $O(1)$ term in the two-constants estimate is then absorbed for all sufficiently large m, giving the exponent used in Lemma 3.6. Finally choose the fixed Esseen fraction $0 < \theta < 1/4$ so that the cubic remainder in the logarithmic expansion is at most one quarter of the Gaussian quadratic damping on $|t| \leq \theta R$.

The choices are made in the order

$$A \to c_0 \to \Gamma \to B \to a_0 \to \gamma_* \to \theta,$$

and no later choice affects an earlier inequality.

B A uniform Laguerre estimate

We prove the envelope used in Lemma 4.1. For $\alpha = -1/2$, introduce the normalized Laguerre function

$$\mathcal{L}_m^{(-1/2)}(r) = a_m L_m^{(-1/2)}(r) e^{-r/2} r^{-1/4}, \qquad a_m = \sqrt{\frac{m!}{\Gamma(m+1/2)}}, \qquad \nu = 4m+1.$$

Proposition 3.2 of Imekraz, Robert, and Thomann [25] applies for every $\alpha > -1$, hence in particular for $\alpha = -1/2$. It gives constants $C, c > 0$, independent of m and r, such that

$$|\mathcal{L}_m^{(-1/2)}(r)| \le C(r\nu)^{-1/4} \quad (0 < r \le \nu^{-1}),$$

$$|\mathcal{L}_m^{(-1/2)}(r)| \le C(r\nu)^{-1/4} \quad (\nu^{-1} \le r \le \nu/2),$$

$$|\mathcal{L}_m^{(-1/2)}(r)| \le C\nu^{-1/4}\big(\nu^{1/3} + |\nu - r|\big)^{-1/4} \quad (\nu/2 \le r \le 3\nu/2),$$

and

$$|\mathcal{L}_m^{(-1/2)}(r)| \le Ce^{-cr} \quad (r \ge 3\nu/2).$$

Here

$$a_m^{-1} \asymp m^{-1/4}, \qquad \sqrt{m}\, a_m^{-1} \asymp m^{1/4} \asymp \nu^{1/4}.$$

Since

$$L_m^{(-1/2)}(r) = a_m^{-1}\mathcal{L}_m^{(-1/2)}(r)e^{r/2}r^{1/4},$$

we need to bound

$$P_m(r) := \sqrt{m}\, e^{-2r/3}|L_m^{(-1/2)}(r)| = \sqrt{m}\, a_m^{-1} r^{1/4} e^{-r/6}|\mathcal{L}_m^{(-1/2)}(r)|.$$

For $r > 0$ the identities above hold literally. The estimate for $P_m(0)$ follows by continuity after the factors $r^{1/4}$ and $r^{-1/4}$ cancel.

In the first two regions, $0 < r \le \nu/2$, the common estimate gives

$$P_m(r) \le C\nu^{1/4} r^{1/4} (r\nu)^{-1/4} e^{-r/6} \le C.$$

In the turning region, $\nu/2 \le r \le 3\nu/2$,

$$P_m(r) \le Cr^{1/4}\big(\nu^{1/3} + |\nu - r|\big)^{-1/4} e^{-r/6} \le C\nu^{1/6}e^{-\nu/12} \le C.$$

Finally, for $r \ge 3\nu/2$,

$$P_m(r) \le C\nu^{1/4} r^{1/4} e^{-(c+1/6)r} \le C.$$

Thus

$$\sup_{m \ge 2,\, r \ge 0} \sqrt{m}\, e^{-2r/3}|L_m^{(-1/2)}(r)| < \infty.$$

Substituting $r = 3x^2/2$ gives exactly

$$\sup_{m \ge 2,\, x \in \mathbb{R}} \sqrt{m}\, e^{-x^2}|L_m^{(-1/2)}(3x^2/2)| < \infty,$$

which is (9).

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